THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. How many zeros are there in one million written in figures? 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

2. The value of $\frac{1}{10^{-3}}$ is: 1

(a) -1000 **(b)** 1000 **(c)** $\frac{1}{1000}$ **(d)** 30

3. The value of $(-4)^2$ is: 1

(a) 16 **(b)** 18 **(c)** -16 **(d)** 8

4. The standard form of 149600000 is: 1

(a) 1.496×10^9 **(b)** 1.496×10^8 **(c)** 14.96×10^8 **(d)** 1.496×10^7

5. Which of the following equals 1.25×10^6? 1

(a) 0.00000125 **(b)** 1250000 **(c)** 125000 **(d)** 12500000

6. $\frac{1}{100000}$ expressed as a power of 10 is: 1

(a) 10^{-4} **(b)** 10^{-5} **(c)** 10^{-6} **(d)** 10^5

7. In 2^9, the exponent is: 1

(a) 9 **(b)** 2 **(c)** 18 **(d)** 512

8. How many zeros are there in one billion written in figures? 1

(a) 8 **(b)** 11 **(c)** 9 **(d)** 10

9. The value of $\frac{1}{4^{-2}}$ is: 1

(a) -16 **(b)** 8 **(c)** $\frac{1}{16}$ **(d)** 16

10. The value of $(-7)^2$ is: 1

(a) 51 **(b)** 49 **(c)** 14 **(d)** -49

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. India's population is about 1400000000. Express it in standard form. 2

12. Express $(7^3)^2 \times 7^{-4}$ as a single power of 7. 2

13. Write $\left(\frac{3}{10}\right)^{-2}$ in simplest form. 2

14. The diameter of a red blood cell is 0.0000075 m. Express it in standard form. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Express each as a single power of 10: 3

(i) $10^{6} \times 10^{-9}$

(ii) $(10^{2})^{-4}$

(iii) $10^{3} \div 10^{-5}$.

16. Find the value of $\left(\frac{4}{1}\right)^{-1} \times \left(\frac{4}{1}\right)^{-1}$. 3

17. Express the following in standard form: 3

(i) 45000000

(ii) 0.00000078

(iii) 2300.

OR

Express each as a single power of 4: 3

(i) $4^{3} \times 4^{-5}$

(ii) $(4^{1})^{-3}$

(iii) $4^{2} \div 4^{-4}$.

18. Find the value of $\left(\frac{3}{5}\right)^{-1} \times \left(\frac{3}{5}\right)^{-1}$. 3

OR

Express the following in standard form: 3

(i) 186000

(ii) 0.0000031

(iii) 72000000.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find x if 5

(i) $2^{x} \times 2^{5} = 2^{13}$

(ii) $(2^{x})^{6} = 2^{18}$

(iii) $2^{x-6} = \frac{1}{2^{4}}$.

OR

Write 72000 and 0.012 in standard form. Then find their product and quotient in standard form. 5

20. Find x if 5

(i) $10^{x} \times 10^{2} = 10^{7}$

(ii) $(10^{x})^{4} = 10^{28}$

(iii) $10^{x-2} = \frac{1}{10^{5}}$.

OR

Write 0.0005 and 0.000002 in standard form. Then find their product and quotient in standard form. 5

END OF QUESTION PAPER Total: 40 Marks

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